Product Requirements Document (PRD): RealFeel UI

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1. Product Overview & Core Philosophy

Most commercial weather applications display atmospheric data inside a theoretical, shaded meteorological box (the Stevenson screen standard). This app completely flips that paradigm. **RealFeel UI** calculates, calibrates, and visualizes the true thermal stress experienced by a human body on the street by incorporating real-world microclimates, direct solar exposure, absolute moisture, and personal metabolic baselines.

The Foundational Formula

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True Perceived Temp = Base Air Temp + Solar Radiation Premium +
Humidity Friction ± Convective Wind Factor - Microclimate Adjustments
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2. Core Features & System Architecture

Epic 1: The Dual-Index Hero Dashboard

- **Perceived Temperature Dominance:** The prominent visual anchor of the UI is the "True Feel" temperature, adjusted for the user's explicit exposure.
- **The Math Breakdown:** A transparent component displaying the exact environmental premiums. For example: 30°C (Base Air) + 6°C (Sun Premium) + 2°C (Dew Point Friction) = 38°C True Feel.
- **Sweat Efficiency Indicator:** A dedicated widget showing how effectively the body can cool itself via evaporation based on current atmospheric saturation.

Epic 2: Contextual Modifiers (Microclimate Engine)

Users can toggle their immediate environmental attributes to recalculate the apparent index in real-time:

Toggle Category	Input Options	Algorithmic Calculation Impact
Exposure	Sun / Shade / Overcast	Dynamically scales from +0°C (shade) up to +8°C based on

Toggle Category	Input Options	Algorithmic Calculation Impact
		UV Index and Solar Zenith angle.
Environment	Urban (Concrete) / Open / Nature	Factors in the Urban Heat Island (UHI) effect, adding up to +2°C for radiant heat retention from masonry.
Activity Level	Stagnant / Walking / Active	Calibrates current metabolic heat generation variables against street-level convective wind speeds.

Epic 3: Personal Bio-Calibration Settings

To move past generalized "Feels Like" metrics, the configuration matrix must ingest individual biometrics to yield precise physiological comfort predictions:

- **Personal Metabolic Calibration:** Interactive fields for Height, Weight, and customized resting metabolic sliders.
- **Acclimatization Matrix:** A multi-state toggle (*Not Acclimatized / Partially / Fully*) that tracks user adaptation over a rolling 14-day local temperature baseline. A sudden thermal spike flags a "Weather Shock" indicator.
- **Smart Clothing Layer Profile:** Maps real-time outfit selections to fixed *Clo Units* (clothing insulation values) to calculate thermal dissipation capabilities in both extreme heat and deep damp cold.

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3. Secondary MVP Core Screens

1. **Hourly "Safe Window" Timeline:** Replaces standard text forecasts with an interactive 24-hour graph plotting *True Feel* against *Base Air Temp*, explicitly highlighting optimized comfort windows for outdoor activities.
2. **Hyperlocal Thermal Heatmap:** A geographical overlay map displaying microclimates across a city grid, mapping out building shadow-casting vectors, tree-canopy shade, and coastal wind corridors.
3. **Micro-Reporting Crowdsourced Loop:** A dead-simple, single-tap feedback button allowing nearby users to report if conditions feel hotter, cooler, or spot-on, generating an autonomous micro-climate validation loop.

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4. Technical & Data Sourcing Framework

The product can be powered cleanly using deterministic public meteorological feeds (e.g., Open-Meteo, Copernicus, NOAA) mapped to static formulas without initial AI dependency. Advanced feature layers integrate specific technology tiers:

- **Public Data Architecture:** Directly tracks Air Temperature, Wind Speed (reduced to a 2-meter street coefficient), and **Dew Point** (relying on absolute moisture thresholds over deceptive relative humidity percentages).
- **AI Optimization Layer (Computer Vision):** Processes 3D building canopy data to calculate dynamic shadow-casting for street-level walking directions.
- **AI Optimization Layer (Natural Language):** An LLM utility processing technical meteorological arrays into highly stylized, human-readable copy on the terminal interface.